Dayoon Kwon

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Education	Ph.D. Epidemiology, University of California, Los Angeles	06/2025
	M.P.H. Epidemiology, New York University	05/2018
	B.S. Biology *with Honors*, Purdue University	05/2016

Publications (28 peer-reviewed, 12 first-author) Google Scholar

1. **Kwon D**, Paul KC, Lin Y, Thordarson DN, Wu J, Su J, Jones DP, Bronstein JM, Ritz B. Untargeted serum metabolomics and air pollution in Parkinson's disease. *npj Parkinson's Disease*. 2026;12(1):244.
2. **Kwon D**, Shao K, O'Sharkey K, Yu Y, Su J, Bronstein JM, Ritz B. Traffic-related air toxics and Parkinson's disease in California's Central Valley. *Science of the Total Environment*. 2026;1046:181999.
3. **Kwon D**, Bozack AK, Ferreccio C, McCormick N, Steinmaus CM, de la Rosa R, Cardenas A. Arsenic exposure at birth, socioeconomic status, and epigenetic aging among adults in northern Chile. *Environmental Research*. 2026;299:124377.
4. Zhang K, Paul KC, Jacobs JP, Lin Y, **Kwon D**, Nianogo R, Bronstein JM, Keener AM, Yu Y, Folle AD, Jones DP, Ritz B. Integration of the serum metabolome and gut microbiome underscores the importance of altered lipid metabolism and potential immune modulation in Parkinson's Disease. *Brain Disorders*. 2026;21:100310.
5. Fox MM, Bramen JE, **Kwon D**, Espeland MA, Sun Y, Zhang L, Shadyab AH, Casanova R, Manson JE, Chang MY, Porter VR, Farland LV, Xiao Q, Wild RA, Siddarth P, Crandall CJ. Pregnancy and breastfeeding are associated with less later-life cognitive decline in a longitudinal, prospective cohort. *Alzheimer's & Dementia*. 2026;22(1):e71072.
6. **Kwon D**, Fessler DMT, Knorr DA, Wiley KS, Sartori J, Coall DA, Fox MM. Of scents and cytokines: How olfactory and food aversions relate to nausea and immunomodulation in early pregnancy. *Evolution, Medicine, and Public Health*. 2025;13(1):269-280.
7. **Kwon D**, Paul KC, Kusters C, Wu J, Bronstein JM, Lill CM, Ketzel M, Raaschou-Nielsen O, Hansen J, Ritz B. Interaction Between Traffic-Related Air Pollution and Parkinson Disease Polygenic Risk Score. *JAMA Network Open*. 2025;8(3):e250854.
8. **Kwon D**, Paul KC, O'Sharkey K, Paik SA, Yu Y, Bronstein JM, Ritz B. Challenges in studying air pollution to neurodegenerative diseases. *Environmental Research*. 2025;278:121597.
9. Ying K, Paulson S, Eames A, Tyshkovskiy A, Li S, Eynon N, Jacques M, Grolaux R, Seale K, Jacques E, Goeminne LJE, Cipriano A, Perez-Guevara M, Emamifar M, Casas Martínez M, **Kwon D**, Kosheleva A, Snyder M, Gobel D, Herzog C, McCartney DL, Marioni RE, Lasky-Su J, Poganik JR, Moqri M, Gladyshev VN. A unified framework for systematic curation and evaluation of aging biomarkers. *Nature Aging*. 2025;5(11):2323-2339.
10. Wiley KS, Martínez LE, **Kwon D**, Knorr DA, Epeldegui M, Fox MM. Regulatory B-Cells Are Associated Negatively With Regulatory T-Cells and Positively With Cytokines in Peripheral Blood of Pregnant Women. *American Journal of Reproductive Immunology*. 2025;93(2):e70027.

11. Fox MM, Hassan A, Wiley KS, **Kwon D**, Knorr DA. Regulatory T-Cells During Pregnancy Relate to Women's Own Childhood History of Microbial Exposure. *American Journal of Human Biology*. 2025;37(3):e70013.
12. Cardoso BR, Liu J, Machado P, **Kwon D**, Belsky DW, Martinez Steele E. Association between ultra-processed food intake and biological ageing in US adults: findings from National Health and Nutrition Examination Survey (NHANES) 2003-2010. *Age and Ageing*. 2024;53(12):afae268.
13. **Kwon D**, Zhang K, Paul KC, Folle AD, Del Rosario I, Jacobs JP, Keener AM, Bronstein JM, Ritz B. Diet and the gut microbiome in patients with Parkinson's disease. *npj Parkinson's Disease*. 2024;10(1):89.
14. Wiley KS, **Kwon D**, Knorr DA, Fox MM. Regulatory T-cell phenotypes in prenatal psychological distress. *Brain, Behavior, and Immunity*. 2024;116:62-69.
15. **Kwon D**, Paul KC, Yu Y, Zhang K, Folle AD, Wu J, Bronstein JM, Ritz B. Traffic-related air pollution and Parkinson's disease in central California. *Environmental Research*. 2024;240(Pt 1):117434.
16. **Kwon D**, Knorr DA, Wiley KS, Young SL, Fox MM. Association of pica with cortisol and inflammation among Latina pregnant women. *American Journal of Human Biology*. 2024;36(5):e24025.
17. Boen CE, Yang YC, Aiello AE, Dennis AC, Mullan Harris K, **Kwon D**, Belsky DW. Patterns and Life Course Determinants of Black-White Disparities in Biological Age Acceleration: A Decomposition Analysis. *Demography*. 2023;60(6):1815-1841.
18. Fox MM, Knorr DA, **Kwon D**, Wiley KS, Parrish MH. How prenatal cortisol levels relate to grandmother-mother relationships among a cohort of Latina women. *American Journal of Human Biology*. 2023;35(7):e23883.
19. **Kwon D**, Folle AD, Del Rosario I, Zhang K, Paul KC, Keener AM, Bronstein JM, Ritz B. Diet quality and Parkinson's disease: Potential strategies for non-motor symptom management. *Parkinsonism & Related Disorders*. 2023;115:105816.
20. Belsky DW, Caspi A, Corcoran DL, Sugden K, Poulton R, Arseneault L, Baccarelli A, Chamarti K, Gao X, Hannon E, Harrington HL, Houts R, Kothari M, **Kwon D**, Mill J, Schwartz J, Vokonas P, Wang C, Williams BS, Moffitt TE. DunedinPACE, a DNA methylation biomarker of the pace of aging. *eLife*. 2022;11:e73420.
21. Erdenetuya B, **Kwon D**, Choi S, Choe SA. Alcohol use during pregnancy: findings from a gender-based violence survey in Mongolia. *Archives of Women's Mental Health*. 2022;25(4):789-795.
22. Graf GH, Crowe CL, Kothari M, **Kwon D**, Manly JJ, Turney IC, Valeri L, Belsky DW. Testing Black-White Disparities in Biological Aging Among Older Adults in the United States: Analysis of DNA-Methylation and Blood-Chemistry Methods. *American Journal of Epidemiology*. 2022;191(4):613-625.
23. Graf GH, Zhang Y, Domingue BW, Harris KM, Kothari M, **Kwon D**, Muennig P, Belsky DW. Social mobility and biological aging among older adults in the United States. *PNAS Nexus*. 2022;1(2):pgac029.
24. **Kwon D**, Choe YJ, Kim SY, Chun BC, Choe SA. Ambient Air Pollution and Kawasaki Disease in Korean Children: A Study of the National Health Insurance Claim Data. *Journal of the American Heart Association*. 2022;11(9):e024092.
25. Graf GH, Li X, **Kwon D**, Belsky DW, Widom CS. Biological aging in maltreated children followed up into middle adulthood. *Psychoneuroendocrinology*. 2022;143:105848.
26. **Kwon D**, Belsky DW. A toolkit for quantification of biological age from blood chemistry and organ function test data: BioAge. *GeroScience*. 2021;43(6):2795-2808.

27. Crowe CL, Domingue BW, Graf GH, Keyes KM, **Kwon D**, Belsky DW. Associations of Loneliness and Social Isolation With Health Span and Life Span in the U.S. Health and Retirement Study. *Journals of Gerontology: Series A*. 2021;76(11):1997-2006.
28. Belsky DW, Caspi A, Arseneault L, Baccarelli A, Corcoran DL, Gao X, Hannon E, Harrington HL, Rasmussen LJH, Houts R, Huffman K, Kraus WE, **Kwon D**, Mill J, Pieper CF, Prinz JA, Poulton R, Schwartz J, Sugden K, Vokonas P, Williams BS, Moffitt TE. Quantification of the pace of biological aging in humans through a blood test, the DunedinPoAm DNA methylation algorithm. *eLife*. 2020;9:e54870.

Working Papers

29. Klingler MC, **Kwon D**, Knorr DA, Wiley KS, Non AL, Fox MM. Social relationships, psychological distress, and maternal immune function during pregnancy. *Submitted*.
30. Rabinowitz J, Green O, **Kwon D**, Burak N, Darawshi M, Belsky DW. Accelerated biological aging linked to high normal serum sodium in general healthcare electronic medical records and NHANES. *Submitted*.
31. **Kwon D**, Khodasevich D, Shadyab AH, Franceschini N, Cardenas A. Epigenome-wide association study of cardiovascular disease biomarkers in postmenopausal women from the Women's Health Initiative. *In Preparation*.
32. **Kwon D**, Paul KC, Folle AD, Wu J, Su J, Keener AM, Bronstein JM, Ritz B. Traffic-related air pollution and Parkinson's disease progression in central California. *In Preparation*.
33. **Kwon D**, Paul KC, Lin Y, Thordarson D, Su J, Jones DP, Bronstein JM, Ritz B. Air toxics exposure and the serum metabolome in Parkinson's disease. *In Preparation*.
34. Shao K, Kusters CDJ, **Kwon D**, Blackmer-Raynolds L, Bronstein J, Keener AM, Wu J, Su J, Paul KC, Ritz BR. Air Pollution Exposures affect Immune Cells Profiles in Parkinson's Disease. *In Preparation*.

Research Support

Funder: NIH (K99/R00) — *Submitted* 06/2026
Title: Multi-omics Biomarkers of Prenatal Air Pollution and ADHD Risk Across Childhood
Role: Principal Investigator

Research Positions

Postdoctoral Scholar for Dr. Andres Cardenas — 07/2025–Present
Department of Epidemiology & Population Health, Stanford University

Research Assistant for Dr. Beate Ritz — 09/2022–06/2025
Department of Epidemiology, UCLA

Research Assistant for Dr. Molly Fox — 02/2022–06/2025
Department of Anthropology, UCLA

Research Assistant for Dr. Seung-Ah Choe — 05/2021–08/2021
Department of Preventive Medicine, Korea University

Project Coordinator for Dr. Daniel Belsky — 07/2019–06/2020
Department of Epidemiology, Columbia University

Research Assistant for Dr. Jiyoung Ahn — 07/2018–06/2019
Department of Population Health, NYU

Research Assistant for Dr. Elodie Ghedin — 09/2016–05/2018
Center for Genomics & Systems Biology, NYU

Research Assistant for Dr. Daniel Suter — 08/2014–05/2016
Department of Biological Sciences, Purdue University

Research Assistant for Dr. Chan Hee Chae — 05/2014–08/2014
Department of Veterinary Pathology, Seoul National University

Awards, Honors, & Fellowships

Notable Korean Scientists, Biological Research Information Center	03/2025
Outstanding Publication Award Honorable Mention, GeroScience	06/2024
Departmental Fellowship, UCLA	09/2022–06/2024
Teaching Apprentice Service Award, UCLA	01/2023–03/2023
Graduate Global Public Health Scholarship, NYU	08/2016–05/2018
Honors Research in Biology, Purdue University	08/2014–05/2016

Conference Travel Awards: UCLA Department of Epidemiology 05/2023, 05/2024; UCLA Graduate Division 05/2023; International Society for Environmental Epidemiology (ISEE) North America Chapter 05/2023

Presentations

Oral Presentations (*presented by co-author)

ISEE North America Chapter, Baltimore, MD. "Wildfire $PM_{2.5}$ exposure and epigenetic aging in salivary cells in northern California." — 06/2026

International Society of Exposure Science–ISEE, Atlanta, GA. "Untargeted serum metabolomics and air pollution in Parkinson's disease." — 08/2025

ISEE, Santiago, Chile. "Impact of long-term exposure to traffic-related air pollution and genetic susceptibility on Parkinson's disease." — 08/2024

ISEE*, Kaohsiung, Taiwan. "Traffic-related air pollution and Parkinson's disease in central California." — 09/2023

International Society for Evolution, Medicine & Public Health, Irvine, CA. "Association of pica with cortisol and inflammation among Latina pregnant women." — 08/2023

Poster Presentations

American Society of Human Genetics, Montreal, Canada. "Epigenome-wide association study of cardiovascular disease biomarkers in postmenopausal women from the Women's Health Initiative." — 10/2026

World Parkinson Congress*, Barcelona, Spain. "Association of diet with gut microbiome in patients with Parkinson's disease." and "A low-quality diet is associated with Parkinson's disease in central California." — 07/2023

ISEE North America Chapter, Corvallis, OR. "Traffic-related air pollution and Parkinson's disease in central California." — 06/2023

Human Biology Association, Reno, NV. "Association between pica during pregnancy and inflammatory response in Latina women." — 04/2023

Invited Seminars

Korea University, Center for Fine Particulate Matter Research, Seoul, Korea. "Air pollution and Parkinson's disease: gene-environment interaction and metabolomic signatures." — 05/2025

Sookmyung Women's University, College of Pharmacy, Seoul, Korea. "Air pollution and Parkinson's disease: from genetic susceptibility to metabolomic signatures." — 05/2025

Professional Service	Ad Hoc Journal Reviewer (*co-reviewed with advisor) JAMA Neurology* • npj Parkinson's Disease • Neurology* • Movement Disorders • Environmental Pollution • GeroScience	
Teaching Experience	Teaching Assistant, UCLA EPIDEM 200B *Epidemiology Methods II: Prediction and Validity* (Ph.D. course)	01/2023–03/2023
Industry Experience	Project Manager, Global Real-World Evidence Johnson & Johnson, Raritan, New Jersey	08/2024–04/2025
	Intern, Sales & Marketing Zoetis (formerly Pfizer Animal Health), Seoul, Korea	06/2013–08/2013
Technical Skills	R package: **BioAge** – [GitHub repository] Software/Programming Skills: R, Python, SQL, SAS, Stata, SPSS, LaTeX	